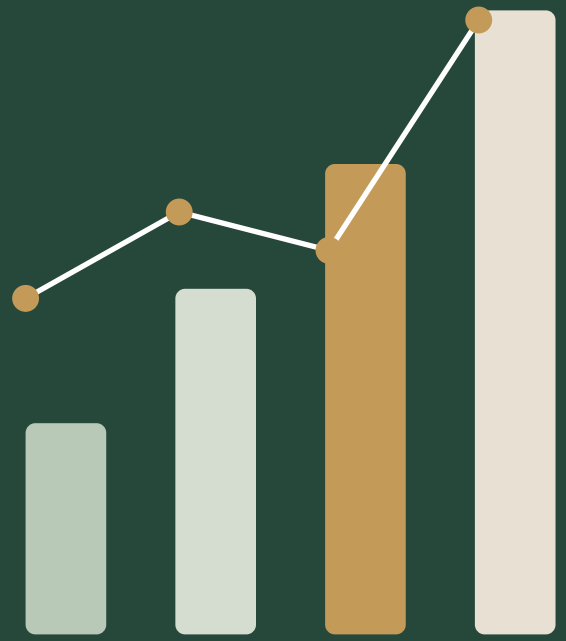
DATA ANALYTICS / PROJECT PRESENTATION

Sales Performance Analysis

Presented by Siraveni Shivani

Information Technology Graduate

Python • Pandas • Excel • Power BI



Exploring sales trends, product performance and profitability

Introduction

Sales analytics turns transaction records into a clearer view of business performance.

UNDERSTAND

Track revenue, profit and changes over time.

COMPARE

Explore products, regions, channels and customers.

ACT

Use evidence to plan growth and allocate resources.

The goal is not just to report what happened, but to make patterns easier to interpret.

Problem Statement

Business question

Where are sales and profit performing differently and what patterns can guide the next decision?

REGIONAL VARIATION

Revenue and profit differ across U.S. regions.

LIMITED VISIBILITY

Seasonality, top SKUs and channel profitability are not immediately clear.

HISTORICAL VIEW

Five years of data can help reveal longer-term patterns.

Analysis focus: trends • profitability • growth opportunities

Project Objectives

01 **Analyze**

Historical sales data

02 **Track**

Monthly and yearly trends

03 **Evaluate**

Product and regional performance

04 **Understand**

Customer behavior and profitability

05 **Build**

An interactive Power BI dashboard

06 **Recommend**

Data-driven business actions

Tools & Technologies

PY

Python

Analysis and preprocessing

PD

Pandas

Data cleaning and manipulation

CO

Google Colab

Notebook-based development

XL

Excel

Inspection and preparation

BI

Power BI

Interactive reporting

MV

Matplotlib / Seaborn

Exploratory visualizations

Dataset Overview

Multiple source tables were consolidated into one analysis-ready dataset.

Sales
Order date, quantity, unit price, revenue, cost

Products
Product name and product identifiers

Budgets
Planning values, including 2017 budget

Customers
Customer name and customer identifiers

Regions + States
State, region and geographic attributes

FINAL DATASET

Identifiers • Financials • Calendar • Geography

Key measures: revenue, cost, profit and profit margin %

Project Workflow

01 **Define objective**

02 **Collect & consolidate**

03 **Load & explore**

04 **Clean & prepare**

05 **Perform EDA**

06 **Build Power BI**

07 **Insights & actions**

A repeatable path from business question to an accessible dashboard.

Data Cleaning & Preprocessing

01 Set the correct header row for the state–region table

02 **Joined sales, customer, product, region, state–region and budget tables**

03 Removed redundant columns and selected analysis fields

04 Standardized column names and renamed fields for clarity

05 **Created profit and profit_margin_pct features**

06 Checked data quality before analysis

Documented data quality check: no missing values or duplicate rows were found.

Exploratory Data Analysis

EDA was used to explore the “what, where and why” behind the sales numbers.

TIME

Monthly and annual sales trends

Line / trend view

PRODUCT

Revenue and profit margin by SKU

Ranked bars

CHANNEL

Sales contribution by channel

Share chart

GEOGRAPHY

State and regional performance

Map / comparison

CUSTOMER

Revenue concentration and margin

Rank / scatter

DRIVERS

Relationships among numeric fields

Correlation heatmap

Sales & Product Performance

MONTHLY CYCLE

Sales commonly fall within a $24M–$26M cycle.

SEASONAL PATTERN

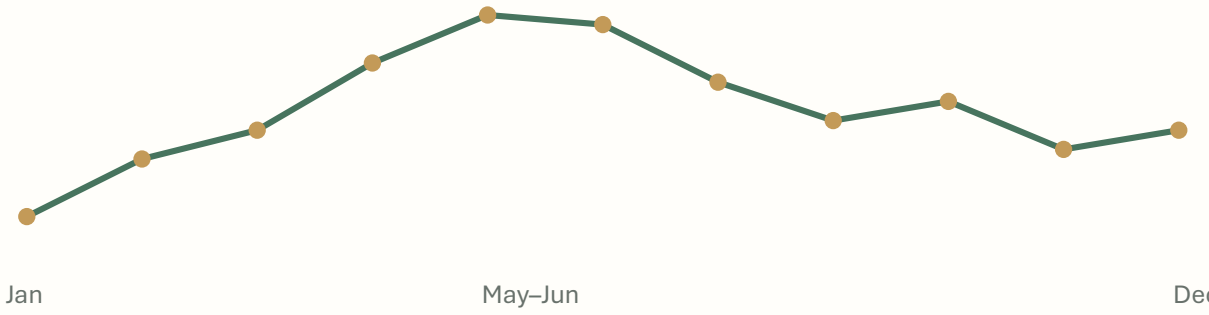
May–June peaks; January is an annual low.

PRODUCT LEADERS

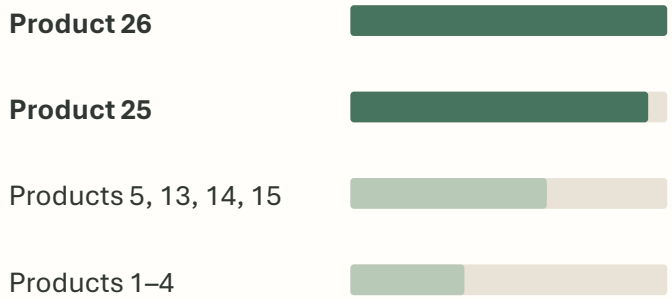
Products 26 and 25 are the top revenue contributors.

MONTHLY SALES TREND

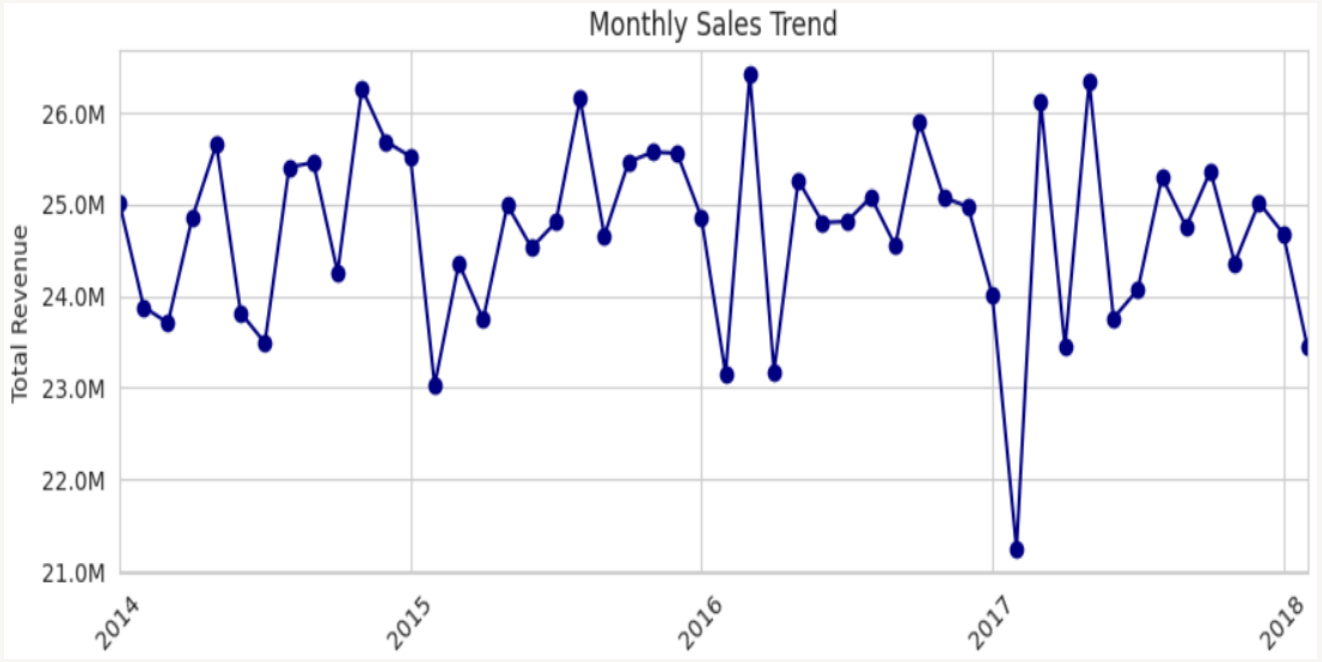
Qualitative pattern from project findings • not to scale

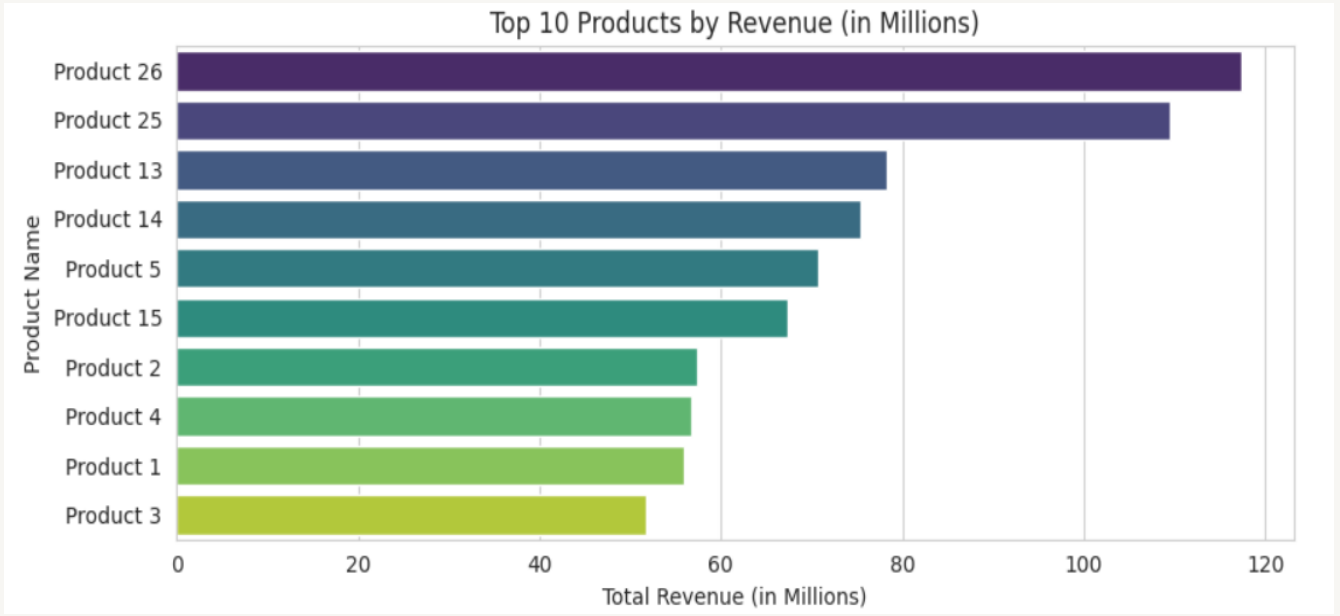


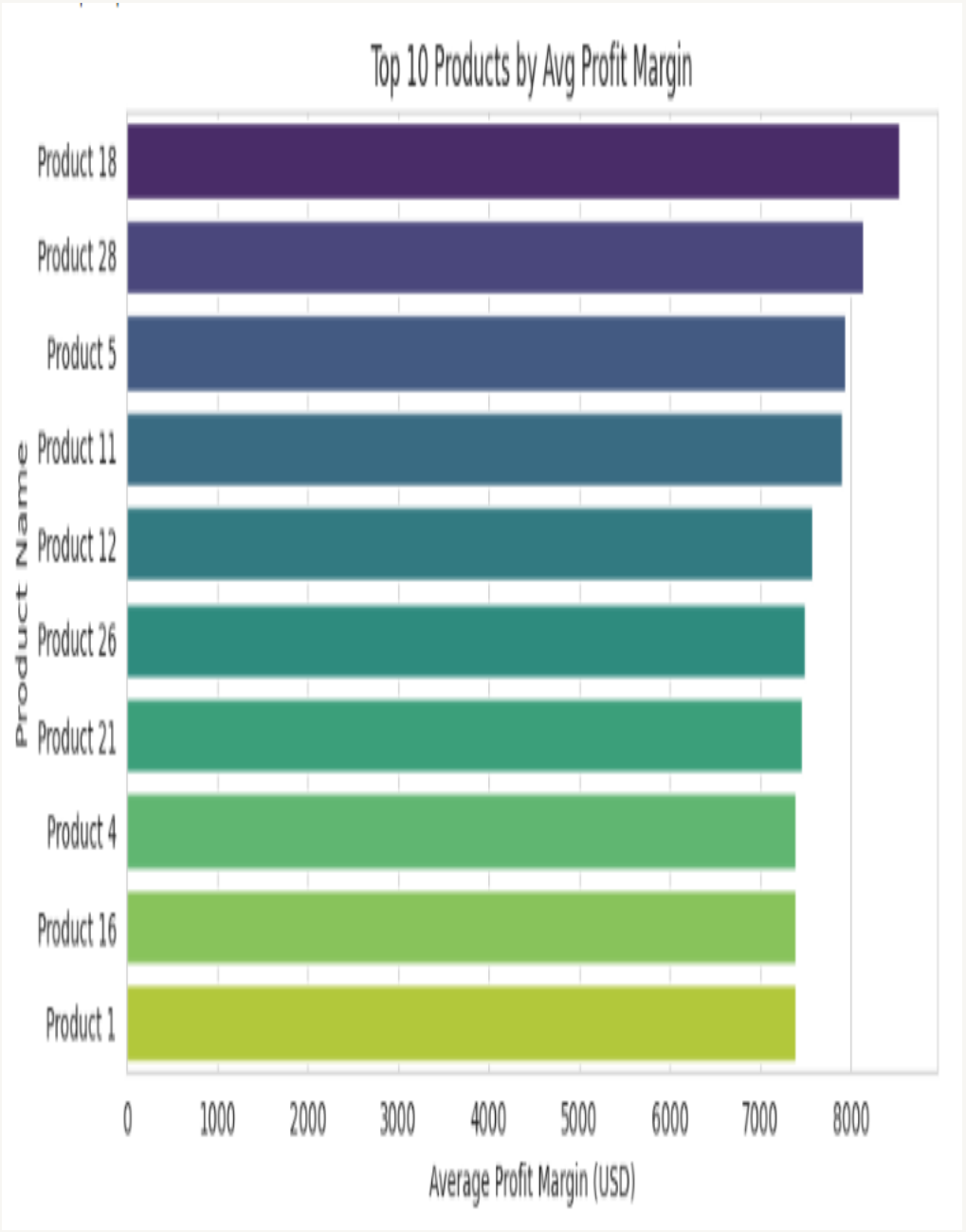
PRODUCT REVENUE RANKING



Ranks reflect documented findings; bar lengths are illustrative.

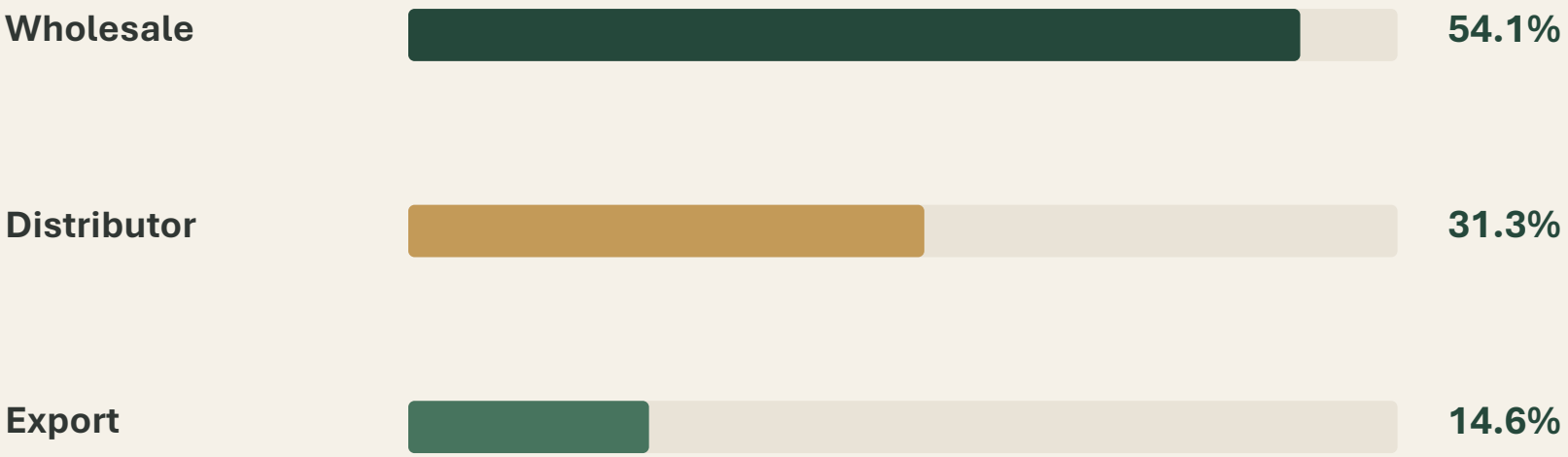

Monthly Sales Trend
Total Revenue
26.0M
25.0M
24.0M
23.0M
22.0M
21.0M
2014
2015
2016
2017
2018


Top 10 Products by Revenue (in Millions)
Product Name
Product 26
Product 25
Product 13
Product 14
Product 5
Product 15
Product 2
Product 4
Product 1
Product 3
0
20
40
60
80
100
120
Total Revenue (in Millions)


Top 10 Products by Avg Profit Margin
Product Name
Product 18
Product 28
Product 5
Product 11
Product 12
Product 26
Product 21
Product 4
Product 16
Product 1
0
1000
2000
3000
4000
5000
6000
7000
8000
Average Profit Margin (USD)

Regional & Channel Analysis

Sales contribution by channel



GEOGRAPHY

West

Highest regional sales

California

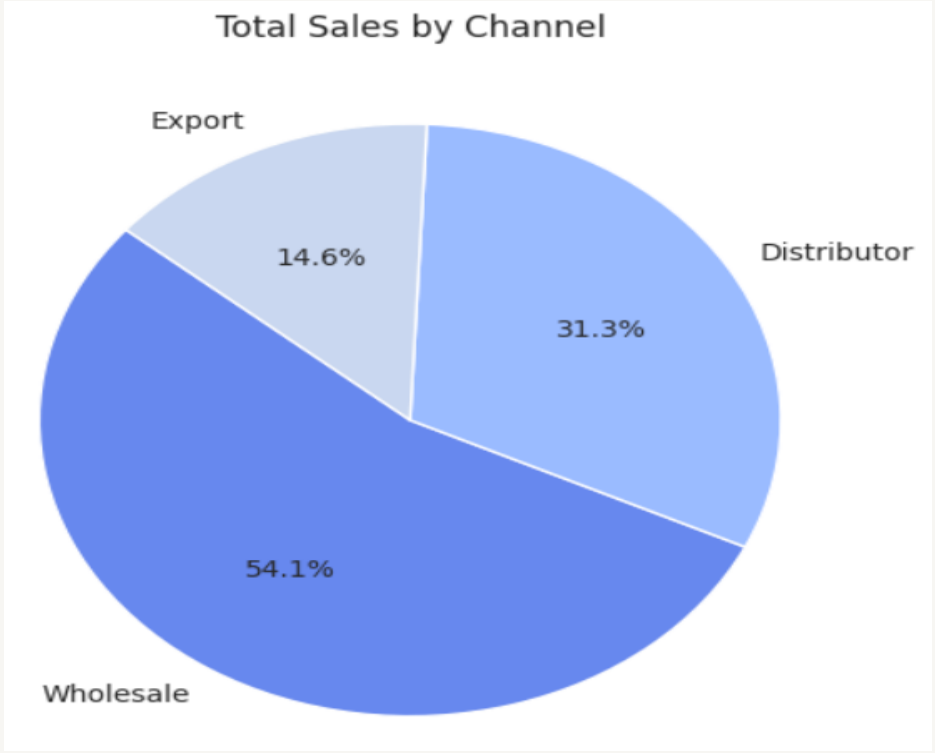
Leads state-level sales

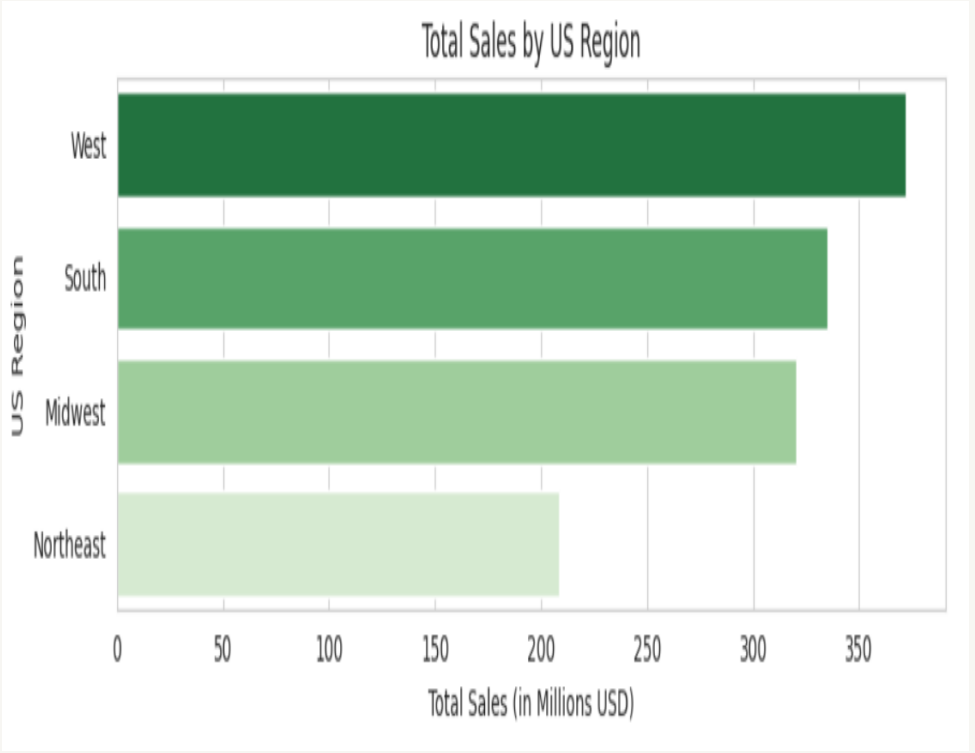
South

Major contributor

Export shows a higher average profit margin than some other channels.

Regional comparisons are based on the project's documented analysis.


Total Sales by Channel
Export
14.6%
Distributor
31.3%
54.1%
Wholesale


Total Sales by US Region
West
South
Midwest
Northeast
US Region
0
50
100
150
200
250
300
350
Total Sales (in Millions USD)

Customer Analysis , Profitability and order values

Original project charts show both account-level and order-level patterns.

REVENUE LEADER

Aibox Company is the top revenue generator.

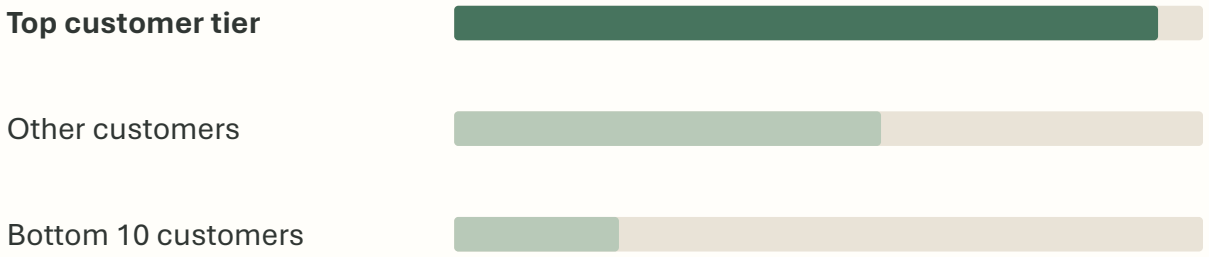
CUSTOMER GAP

Top customers contribute far more than the bottom tier.

MARGIN SIGNAL

Large accounts above $10M with <36% margin are review candidates.

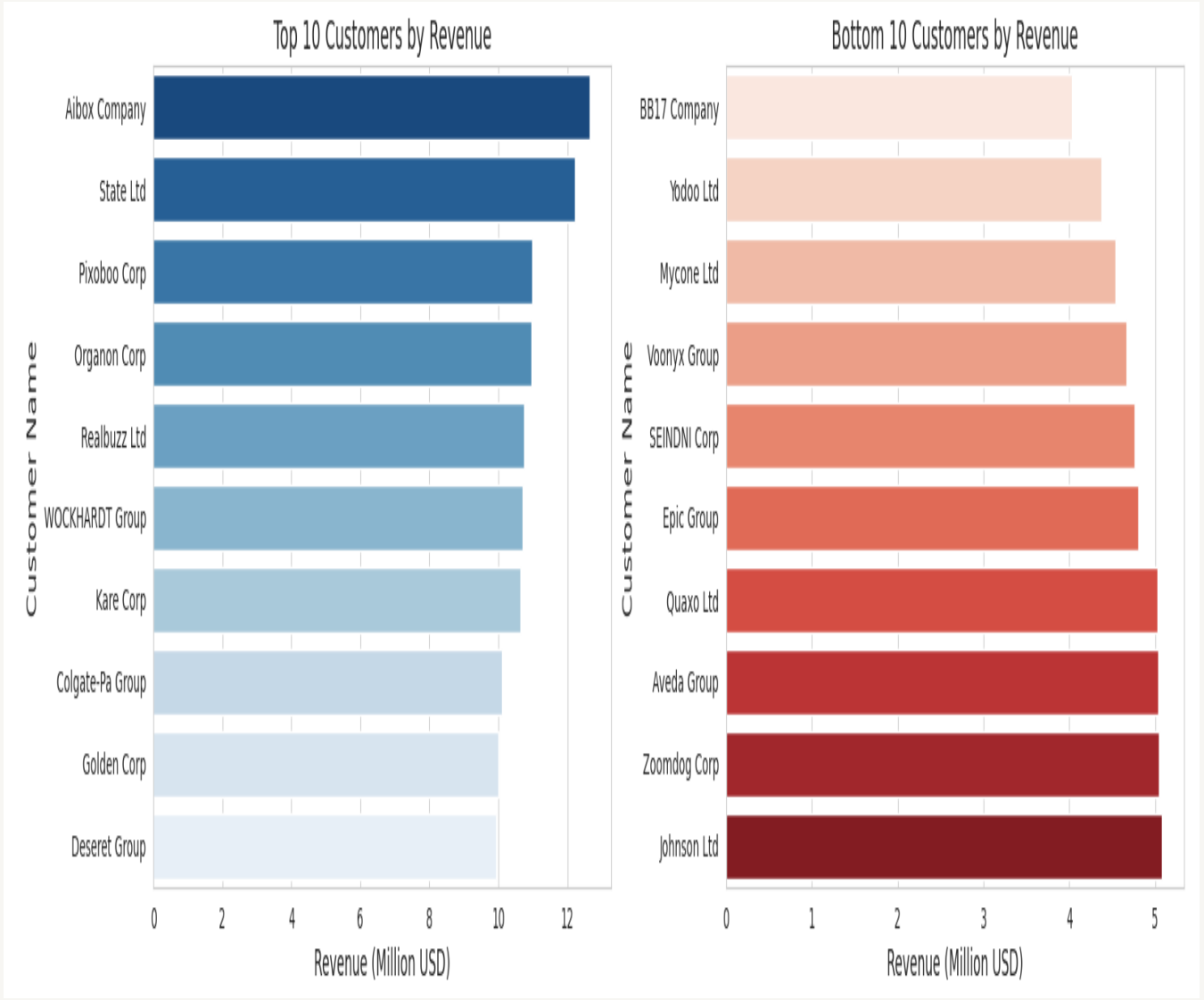
CUSTOMER REVENUE CONCENTRATION

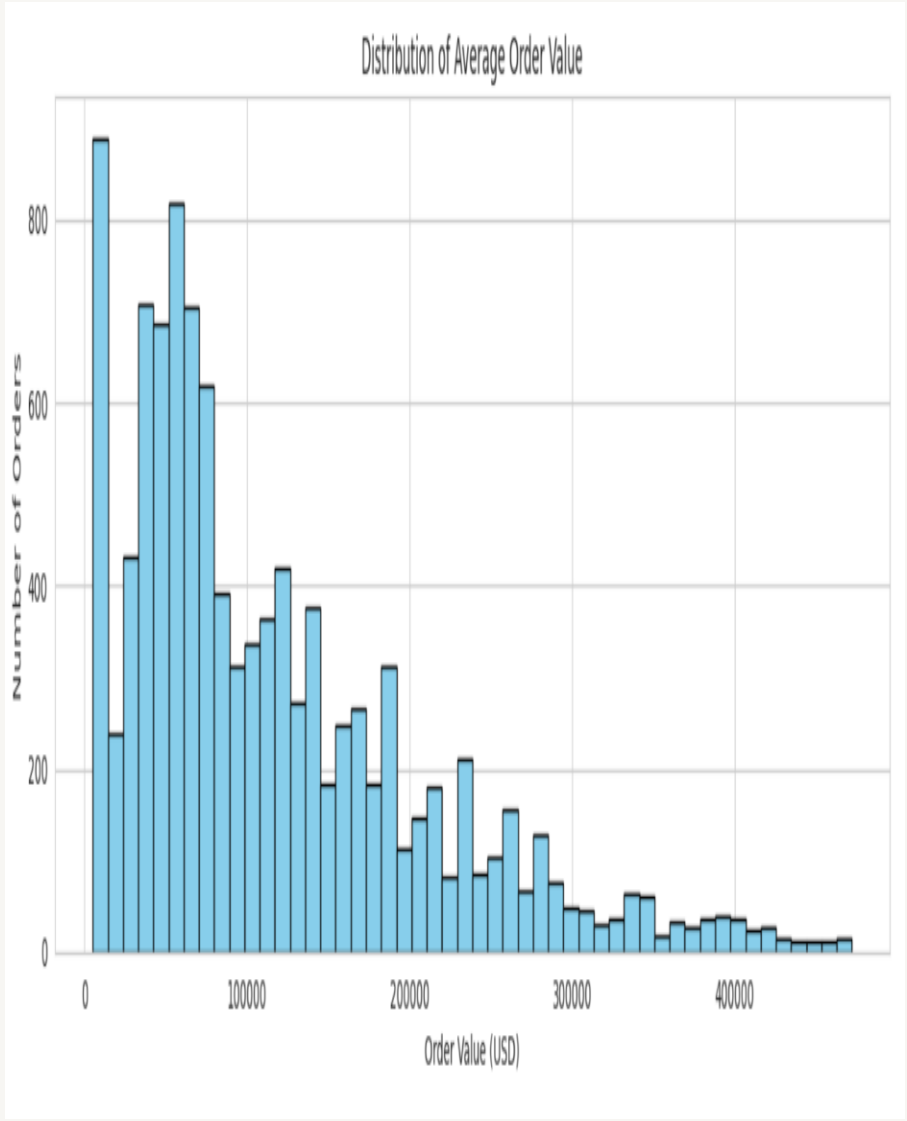


Relative illustration; exact shares not provided.

SEGMENTATION TAKEAWAYS

- >$10M revenue and <36% margin: review terms
- $6–9M revenue and >40% margin: upsell candidates
- Monitor revenue and margin together


Top 10 Customers by Revenue
Aibox Company
State Ltd
Pixoboo Corp
Organon Corp
Realbuzz Ltd
WOCKHARDT Group
Kare Corp
Colgate-Pa Group
Golden Corp
Deseret Group
Customer Name
0
2
4
6
8
10
12
Revenue (Million USD)
Bottom 10 Customers by Revenue
BB17 Company
Yodoo Ltd
Mycone Ltd
Voonyx Group
SEINDNI Corp
Epic Group
Quaxo Ltd
Aveda Group
Zoomdog Corp
Johnson Ltd
Customer Name
0
1
2
3
4
5
Revenue (Million USD)


Distribution of Average Order Value
Number of Orders
0
200
400
600
800
0
100000
200000
300000
400000
Order Value (USD)

Power BI Dashboard

Three dashboard pages support summary reporting and deeper exploration.

Performance Summary



KPIs, monthly trend, channel mix and regional view

Customer Segmentation



Customer revenue, margin and segment comparisons

Revenue Scenarios



Scenario-based revenue exploration

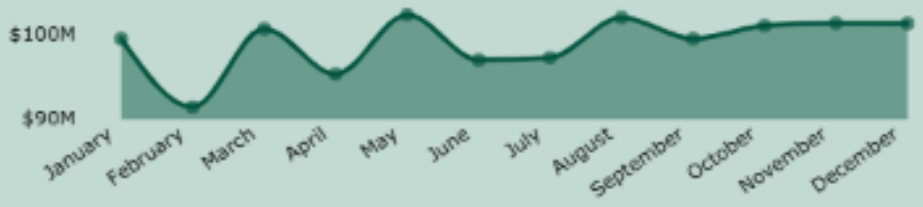
Mockups are illustrative layouts, not screenshots of the original Power BI report.

Executive Overview & Trends | Product & Channel Performance | Geographic & Customer Insights | By Shivani

Total Revenue	Total Profit	Profit Margin %	Total Orders	Revenue per order
$1.2bn	$461.8M	37.36%	64K	$19.3K

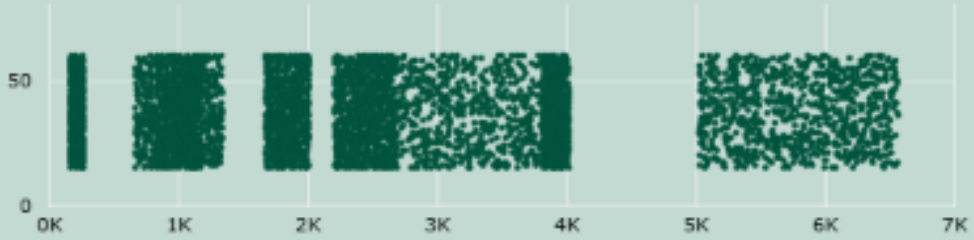
Monthly Revenue Rhythm: Uncovering Seasonality Peaks

Total Revenue by order_month_name



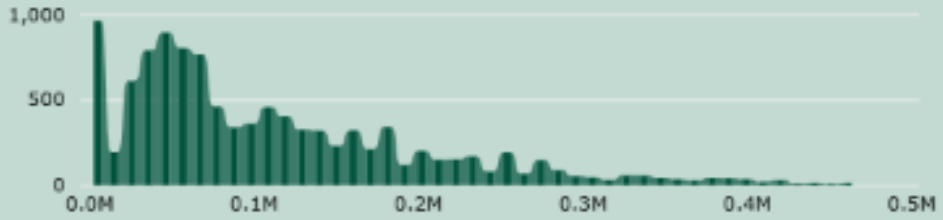
Unit Price Vs Profit Margin: Spotting High Margin Price Bands

unit_price and profit_margin_pct



Order Value Spectrum: Mapping Customer Spending Tiers

Count of order_number by OrderValue (bins)



Profit Pulse: Tracking Monthly Earnings Momentum

Total Profit by order_month_name

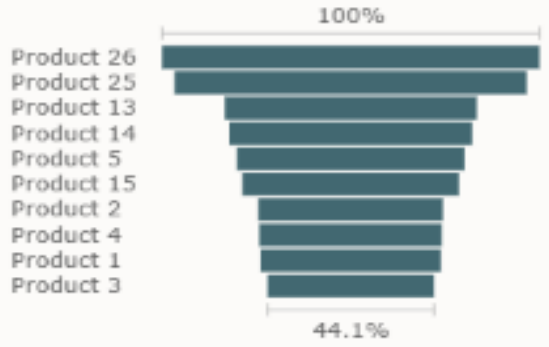


Executive Overview & Trends | Product & Channel Performance | Geographic & Customer Insights | +

Executive Overview & Trends | Product & Channel Performance | Geographic & Customer Insights

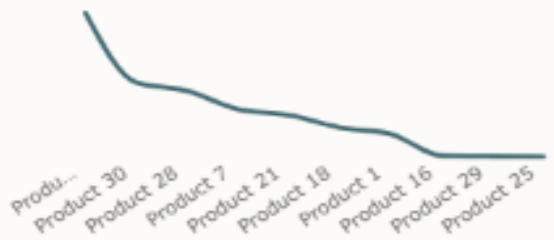
Revenue Champions: Best-Selling Products Driving Growth

Total Revenue by product_name



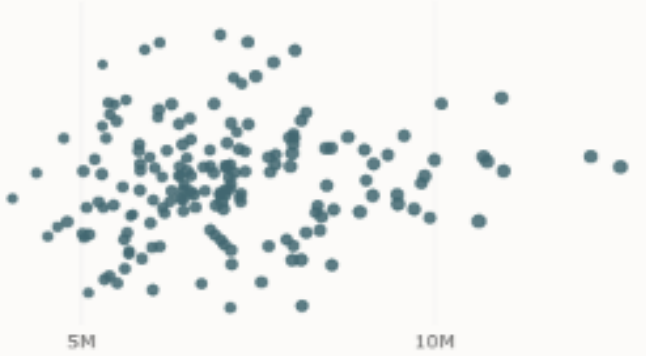
High-Margin Heroes: Most Efficient Products to Sell

Profit Margin % by product_name



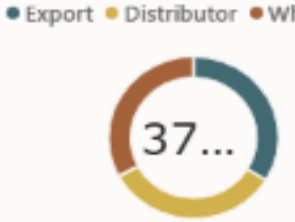
Strategic Product Positioning: Revenue vs. Profitability

Sum of Order Count by Total Revenue and Avg Profit ...



Channel Efficiency Scorecard: Margin per Sale by Route

Profit Margin % by channel



Channel Power Play: Where the Revenue Comes From

Total Revenue by channel



Profit Pipeline by Channel: Who's Really Paying Off?

Total Profit by channel



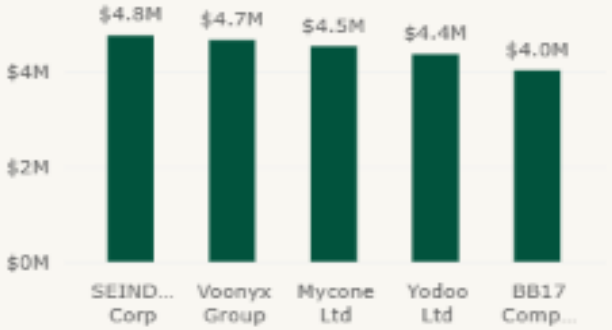
Executive Overview & Trends | Product & Channel Performance | Geographic & Customer Insights

Top 5

Butto...

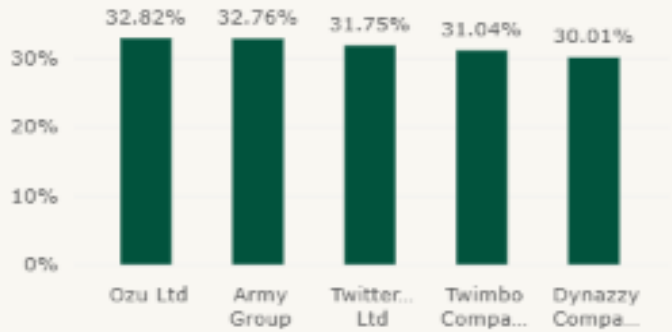
Bottom 5 Customers by Revenue

Total Revenue by customer_name



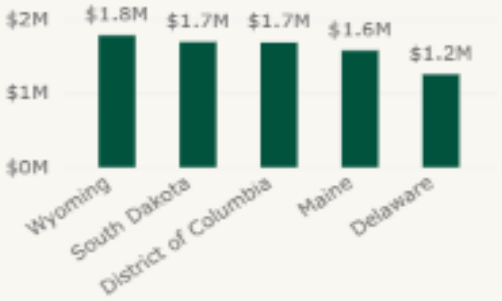
Bottom 5 Customers by Profit Margin

Profit Margin % by customer_name



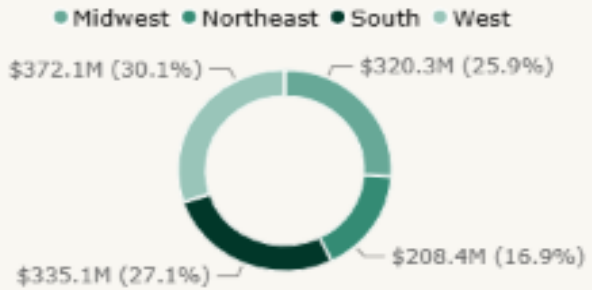
Bottom 5 States by Revenue

Total Revenue by state_name



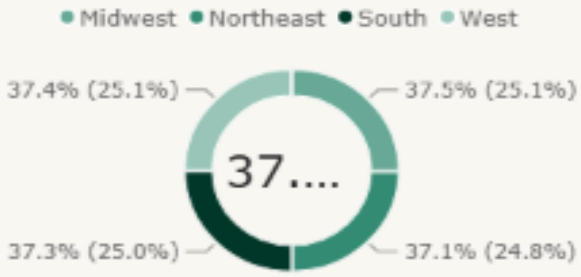
Total Revenue by Region

Total Revenue by us_region



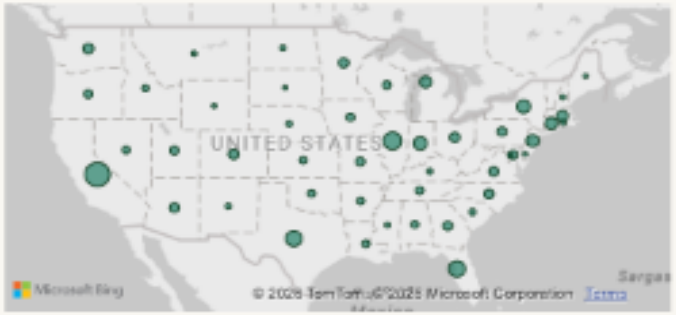
Profit Margin % by Region

Profit Margin % by us_region



Total Profit

by state



Executive Overview & Trends | Product & Channel Performance | Geographic & Customer Insights | +

Key Insights & Recommendations

DOCUMENTED INSIGHTS

- Products 26 and 25 are major revenue contributors
- Wholesale accounts for the largest sales share
- California leads state sales; West leads regions
- Sales show seasonal fluctuations

PRACTICAL RECOMMENDATIONS

- Plan promotions around seasonal lows and peaks
- Prioritize strong products; review low performers
- Develop profitable channel partnerships
- Explore lower-performing regional opportunities

Conclusion

- Completed end-to-end sales data analysis and EDA
- Built an interactive Power BI dashboard
- Identified trends across products, customers, channels and regions
- Developed data-driven recommendations for business planning

THANK YOU

Presented by

Siraveni Shivani